DETERMINING MARKET STRUCTURE

Learning Outcome

- identify the type of market structure within which a firm operates and describe the use and limitations of concentration measures

Monopoly markets and other situations in which companies have pricing power can be inefficient because producers constrain output to cause an increase in prices. Therefore, less of the good will be consumed, and it will be sold at a higher price, which is generally inefficient for the overall market. As a result, many countries have introduced competition law to regulate the degree of competition in many industries.

Market power in the real world is not always as clear as it is in textbook examples. Governments and regulators often have the difficult task of measuring market power and establishing whether a firm has a dominant position that may resemble a monopoly. A few historical examples of this are as follows:

1. In the 1990s, US regulators prosecuted agricultural corporation Archer Daniels Midland for conspiring with Japanese competitors to fix the price of lysine, an amino acid used as an animal feed additive. The antitrust action resulted in a settlement that involved more than US$100 million in fines paid by the cartel members.
2. In the 1970s, US antitrust authorities broke up the local telephone monopoly, leaving AT&T the long-distance business (and opening that business to competitors), and required AT&T to divest itself of the local telephone companies it owned. This antitrust decision brought competition, innovation, and lower prices to the US telephone market.
3. European regulators (specifically, the European Commission) have affected the mergers and monopoly positions of European corporations (as in the case of the companies Roche, Rhone-Poulenc, and BASF, which were at the center of a vitamin price-fixing case) as well as non-European companies (such as Intel) that do business in Europe. Moreover, the merger between the US company General Electric and the European company Honeywell was denied by the European Commission on grounds of excessive market concentration.

Quantifying excessive market concentration is difficult. Sometimes, regulators need to measure whether something that has not yet occurred might generate excessive market power. For example, a merger between two companies might allow the combined company to be a monopolist or quasi-monopolist in a certain market.

A financial analyst hearing news about a possible merger should always consider the impact of competition law (sometimes called antitrust law)—that is, whether a proposed merger may be blocked by regulators in the interest of preserving a competitive market.

Econometric Approaches

How should one measure market power? The theoretical answer is to estimate the elasticity of demand and supply in a market. If demand is very elastic, the market must be very close to perfect competition. If demand is rigid (inelastic), companies *may* have market power.

From the econometric point of view, this estimation requires some attention. The problem is that observed price and quantity are the equilibrium values of price and quantity and do not represent the value of either supply or demand. Technically, this is called the problem of endogeneity, in the sense that the equilibrium price and quantity are jointly determined by the interaction of demand and supply. Therefore, to have an appropriate estimation of demand and supply, we need to use a model with two equations: an equation of demanded quantity (as a function of price, income of the buyers, and other variables) and an equation of supplied quantity (as a function of price, production costs, and other variables). The estimated parameters will then allow us to compute elasticity.

Regression analysis is useful in computing elasticity but requires a large number of observations. Therefore, one may use a time-series approach and, for example, look at 20 years of quarterly sales data for a market. The market structure may have changed radically over those 20 years, however, and the estimated elasticity may not apply to the current situation. Moreover, the supply curve may change because of a merger among large competitors, and the estimation based on past data may not be informative regarding the future state of the market postmerger.

An alternative approach is a cross-sectional regression analysis. Instead of looking at total sales and average prices in a market over time (the time-series approach mentioned earlier), we can look at sales from different companies in the market during the same year, or even at single transactions from many buyers and companies. Clearly, this approach requires a substantial data-gathering effort, and therefore, this estimation method can be complicated. Moreover, different specifications of the explanatory variables (e.g., using total GDP rather than median household income or per-capita GDP to represent income) may lead to dramatically different estimates.

Simpler Measures

this ratio, one would, for example, add the sales values of the largest 10 firms and divide this figure by total market sales. This number is always between 0 (perfect competition) and 100 percent (monopoly).

The main advantage of the concentration ratio is that it is simple to compute, as shown previously. The disadvantage is that it does not directly quantify market power. In other words, is a high concentration ratio a clear signal of monopoly power? A company may be the only incumbent in a market, but if the barriers to entry are low, the simple presence of a *potential* entrant may be sufficient to convince the incumbent to behave like a firm in perfect competition. For example, a sugar wholesaler may be the only one in a country, but the knowledge that other large wholesalers in the food industry might easily add imported sugar to their range of products should convince the sugar wholesaler to price its product as if it were in perfect competition.

Another disadvantage of the concentration ratio is that it tends to be unaffected by mergers among the top market incumbents. For example, if the largest and second-largest incumbents merge, the pricing power of the combined entity is likely to be larger than that of the two preexisting companies. But the concentration ratio may not change much.

Calculating the Concentration Ratio

Suppose there are eight producers of a certain good in a market. The largest producer has 35 percent of the market, the second largest has 25 percent, the third has 20 percent, the fourth has 10 percent, and the remaining four have 2.5 percent each. If we computed the concentration ratio of the top three producers, it would be 35 + 25 + 20 = 80 percent, while the concentration ratio of the top four producers would be 35 + 25 + 20 + 10 = 90 percent.

If the two largest companies merged, the new concentration ratio for the top three producers would be 60 (the sum of the market shares of the merged companies) + 20 + 10 = 90 percent, and the concentration ratio for the four top producers would be 92.5 percent. Therefore, this merger affects the concentration ratio very mildly, even though it creates a substantial entity that controls 60 percent of the market.

For example, the effect of consolidation in the US retail gasoline market has resulted in increasing degrees of concentration. In 1992, the top four companies in the US retail gasoline market shared 33 percent of the market. By 2001, the top four companies controlled 78 percent of the market (Exxon Mobil 24 percent, Shell 20 percent, BP/Amoco/Arco 18 percent, and Chevron/Texaco 16 percent).

To avoid the known issues with concentration ratios, economists O. C. Herfindahl and A. O. Hirschman suggested an index in which the market shares of the top *N* companies are first squared and then added. If one firm controls the whole market (a monopoly), the Herfindahl–Hirschman index (HHI) equals 1. If there are *M* firms in the industry with equal market shares, then the HHI equals (1/*M*). This provides a useful gauge for interpreting an HHI. For example, an HHI of 0.20 would be analogous to having the market shared equally by five firms.

The HHI for the top three companies in the example in the box above would be $0.35^2 + 0.25^2 + 0.20^2 = 0.225$ before the merger, whereas after the merger, it would be $0.60^2 + 0.20^2 + 0.10^2 = 0.410$, which is substantially higher than the initial 0.225. The HHI is widely used by competition regulators; however, just like the concentration ratio, the HHI does not take the possibility of entry into account, nor does it consider the elasticity of demand. Therefore, the HHI has limited use for a financial analyst trying to estimate the potential profitability of a company or group of companies.

EXAMPLE 4

The Herfindahl–Hirschman Index

1. Suppose a market has 10 suppliers, each of them with 10 percent of the market. What are the concentration ratio and the HHI of the top four firms?

   A. Concentration ratio 4 percent and HHI 40
   B. Concentration ratio 40 percent and HHI 0.4
   C. Concentration ratio 40 percent and HHI 0.04

   Hide Solution

   Solution:

   C is correct. The concentration ratio for the top four firms is 10 + 10 + 10 + 10 = 40 percent, and the HHI is $0.10^2 \times 4 = 0.01 \times 4 = 0.04$.

Question Set

Company	Sales (in millions of euros)
ABC	300
Brown	250
Coral	200
Delta	150
Erie	100
All others	50

2. The industry's four-company concentration ratio is *closest* to:

A. 71%.
B. 86%.
C. 95%.

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Solution:

B is correct. The top four companies in the industry account for 86 percent of industry sales: (300 + 250 + 200 + 150)/(300 + 250 + 200 + 150 + 100 + 50) = 900/1050 = 86%.

The following information applies to questions 3 & 4

An analyst gathered the following market share data for an industry composed of five companies:

Company	Market Share (%)
Zeta	35
Yusef	25
Xenon	20
Waters	10
Vlastos	10

3. The industry's three-firm Herfindahl–Hirschman index is *closest* to:

A. 0.185.
B. 0.225.
C. 0.235.

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Solution:

B is correct. The three-firm Herfindahl–Hirschman index is $0.35^2 + 0.25^2 + 0.20^2 = 0.225$.

4. One disadvantage of the Herfindahl–Hirschman index is that the index:

A. is difficult to compute.
B. fails to reflect low barriers to entry.
C. fails to reflect the effect of mergers in the industry.

Hide Solution

Solution:

B is correct. The Herfindahl–Hirschman index does not reflect low barriers to entry that may restrict the market power of companies currently in the market.

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